

Fashion Week Diversity

Does POC model representation differ significantly across the Big Four fashion capitals?

New York . London . Paris . Milan . 2015-2023 . One-Way ANOVA with Bonferroni post-hoc correction

F(3, 32) = 4.58	p = 0.009	Effect size $\eta^2 = 0.30$
ANOVA result	significant at $\alpha = 0.05$	large effect

Background

Runway diversity has been one of the most talked-about issues in fashion over the past decade. But beyond the headlines, is the change statistically meaningful, and is it happening at the same pace across cities? This analysis uses season-level data from The Fashion Spot's annual diversity reports (2015-2023), tracking POC model appearances across New York, London, Paris, and Milan. A one-way ANOVA tests whether mean POC representation differs significantly by city, with Bonferroni-corrected pairwise t-tests identifying exactly which pairs diverge.

Data Summary

City	Seasons	Mean POC %	SD	2015 to 2023
New York	9	40.5%	7.9	28.0% to 50.1%
London	9	35.8%	8.1	24.1% to 46.8%
Paris	9	31.8%	7.3	21.5% to 41.3%
Milan	9	27.3%	8.3	15.3% to 38.7%

Source: The Fashion Spot Annual Runway Diversity Reports. $n = 36$ total (9 seasons x 4 cities).

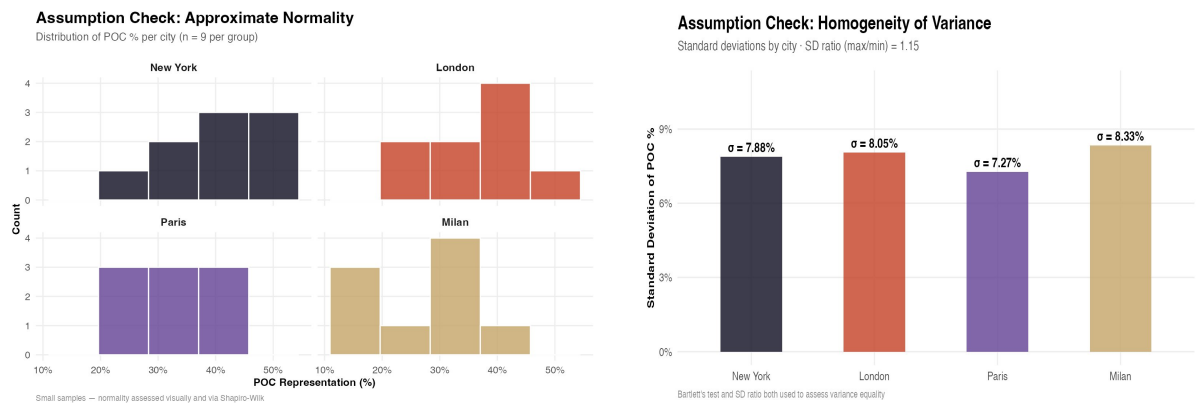
Assumption Checks

Three conditions were verified before running the ANOVA:

Independence - each observation is a distinct city-season pair. Cities are independent fashion markets with non-overlapping show schedules.

Normality - Shapiro-Wilk tests returned $p > 0.05$ for all four cities (range 0.39 to 0.60). No significant departure from normality.

Equal variances - Bartlett's test: $K2(3) = 0.15$, $p = 0.99$. SD ratio (max/min) = 1.15, well under the threshold of 2.



Left: histograms confirm approximate normality per city. Right: near-equal SDs confirm homogeneity of variance.

ANOVA Results

H0: Mean POC representation is equal across all four cities.

H1: At least one city differs significantly.

The one-way ANOVA returned $F(3, 32) = 4.58$, $p = 0.009$. We reject H0 at $\alpha = 0.05$. The effect size $\eta^2 = 0.30$ means roughly 30% of the variance in POC representation is explained by city alone, which is a large and practically meaningful effect.

Distribution of POC Representation by Fashion Capital

Boxplots with individual seasons shown · Diamond = group mean · 2015–2023

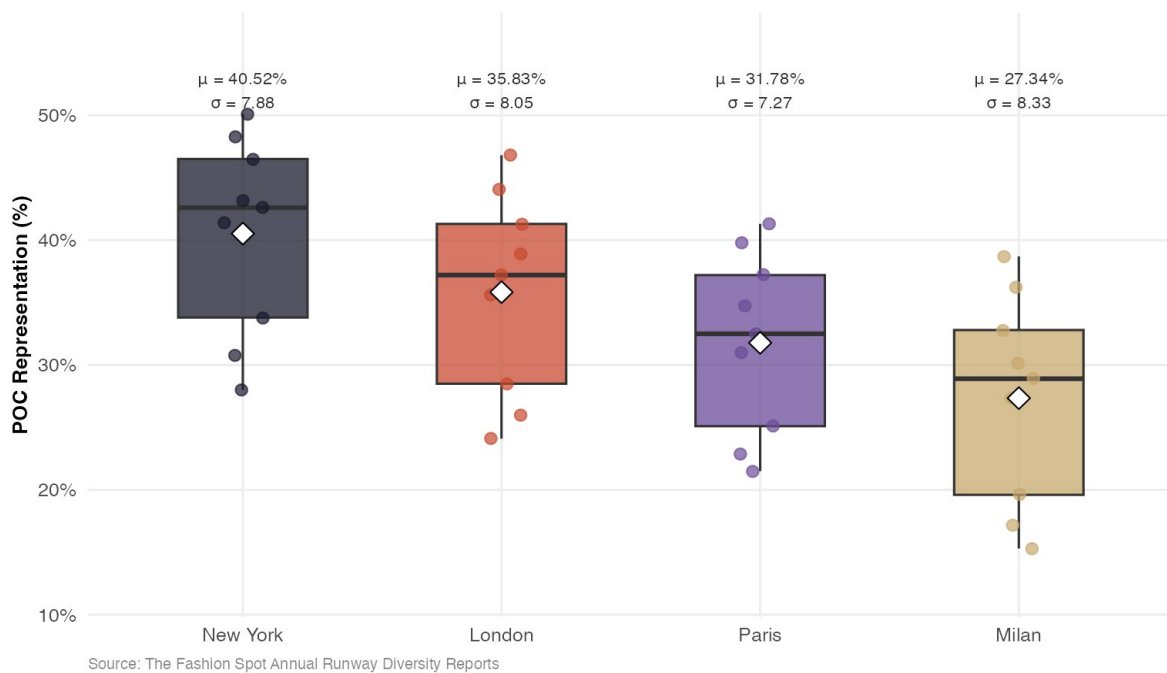
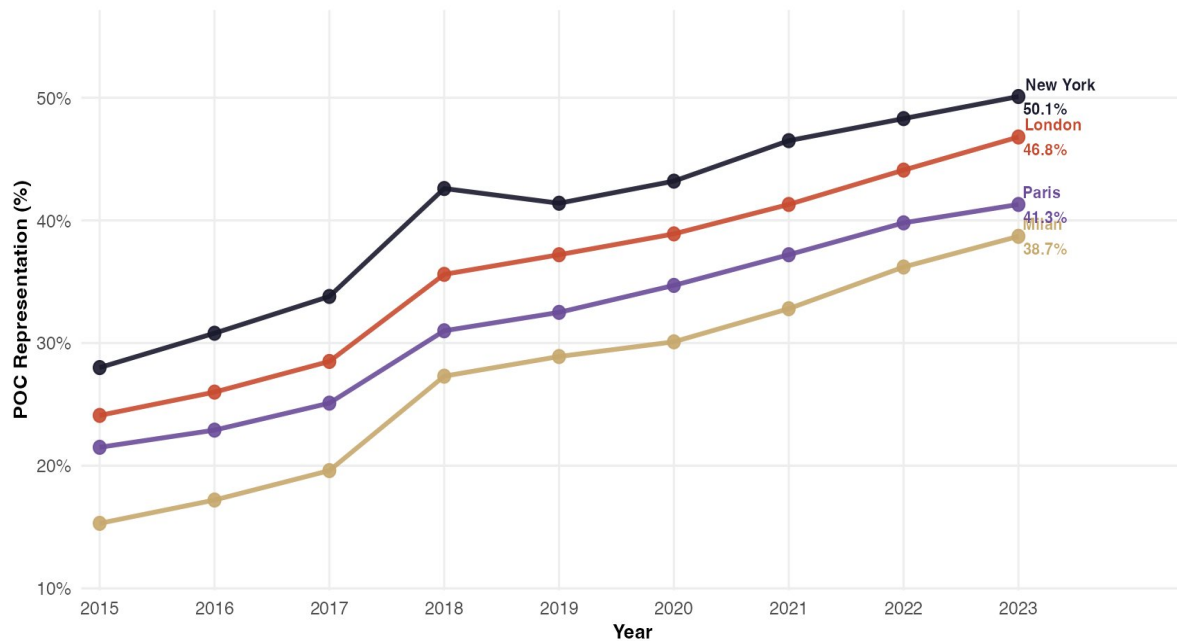


Figure 1. Distributions by city. Diamond = group mean. New York leads; Milan consistently trails.

POC Model Representation at the Big Four Fashion Weeks

Percentage of model appearances by non-white models, 2015–2023



Source: The Fashion Spot Annual Runway Diversity Reports

Figure 2. POC representation over time. All cities improved, but the gaps between them persisted.

Post-Hoc Analysis

With 4 groups there are $C(4,2) = 6$ pairwise comparisons. Bonferroni correction sets the adjusted threshold at $\alpha = 0.05/6 = 0.0083$. Only one pair was significant after correction: New York vs. Milan ($p = 0.007$, mean difference = 13.2 percentage points). All other pairs were non-significant, meaning the overall ANOVA signal is largely driven by the gap between these two cities.

Post-Hoc Pairwise Comparisons (Bonferroni Correction)

Adjusted p-values for all 6 city pairs · *** p<.001 ** p<.01 * p<.05



Figure 3. Bonferroni-corrected p-values for all 6 city pairs. Only NY vs Milan is significant.

Diversity Progress Since 2015 by Fashion Capital

Change in POC representation (percentage points) relative to 2015 baseline

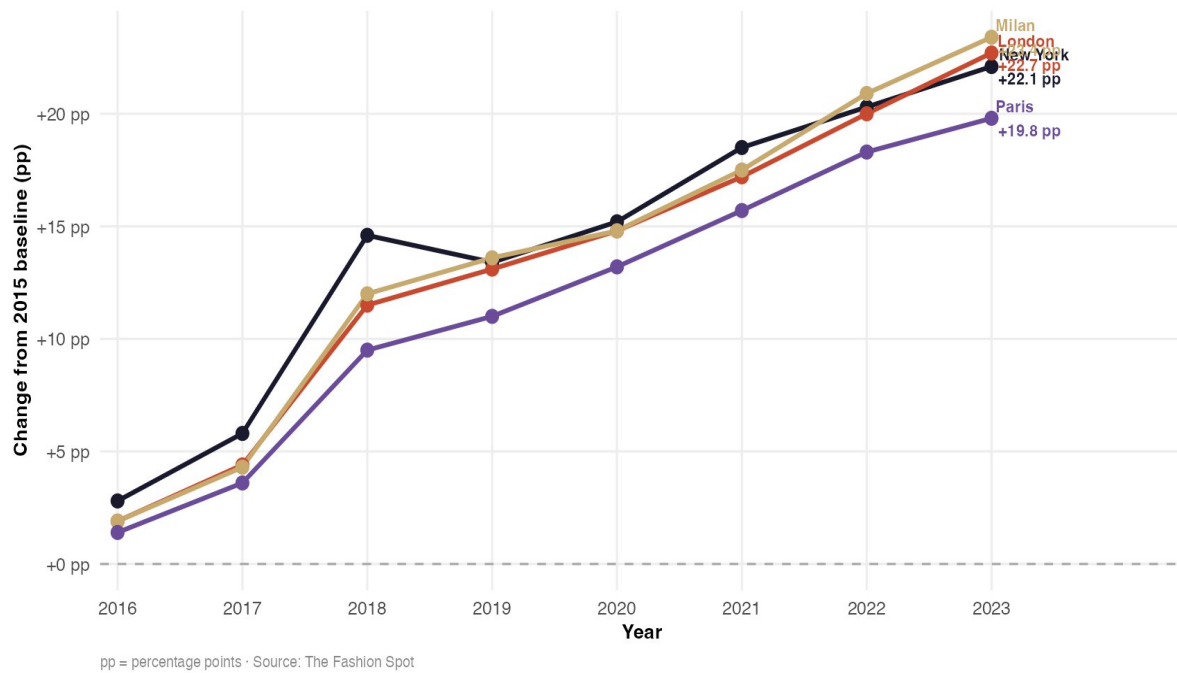


Figure 4. Gains in percentage points relative to the 2015 baseline.

Interpretation

The results confirm that POC representation differs significantly across the Big Four ($F(3,32) = 4.58$, $p = 0.009$, $n^2 = 0.30$). The post-hoc analysis shows the divide is clearest between New York and Milan, which makes sense. Milan has historically leaned into classical luxury aesthetics, making it structurally slower to diversify than New York, which operates in a far more multicultural environment.

Every city improved since 2015, which is a real and measurable shift. But progress accelerated sharply around 2018, right when social media scrutiny of runway casting hit a peak, which raises the question of how much of this reflects genuine cultural change versus image management.

The runway is the visible part. The more interesting question is what is happening behind it: in design teams, in leadership, in supply chains. Those are harder to measure, and probably more telling.

Limitations

- Small sample size ($n = 9$ per city) means results should be interpreted with some caution.
- Season-level aggregates don't capture within-season variation across individual shows.
- POC is a single-dimension measure and doesn't distinguish between ethnic groups.
- Year-to-year trends may carry autocorrelation, which one-way ANOVA does not account for.

Data: The Fashion Spot Annual Runway Diversity Reports (2015-2023) . Analysis: R 4.5.1 . Method: One-Way ANOVA, Bonferroni post-hoc correction